

# PREDICTION IS NOT PLANNING: GEOMETRIC ADVANTAGE RANKING FOR LANGUAGE-ACTION JEPAS

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## ABSTRACT

Agents that act through language must compare consequences, not only generate plausible text. We study action-conditioned joint-embedding prediction in a controlled language-mediated planning environment. We first formalize a planning state as an equivalence class of histories with identical controlled futures. This predictive state is sufficient for control, but its geometry is not identified by transition prediction. Equally predictive re-embeddings can reverse which successor appears closest to a goal. We introduce Geometric Advantage Ranking (GAR), a training objective that imposes within-state order constraints on counterfactual action consequences. GAR can supervise a scorer that factors through a predicted successor or a matched direct scorer. On stylized arithmetic dependency puzzles, a non-generative JEPA solves 43.3% of episodes in the minimum number of actions under a common feasible-action interface. The strongest sentence model reaches 52.1%, so our result establishes viability rather than superiority. The central finding concerns mechanism. Latent prediction alone reaches 18.3%, while the complete predictive GAR model reaches 43.3%. Five-seed ablations show that counterfactual ordering and successor prediction contribute complementary information. Local action-order diagnostics are more favorable for the learned GAR score than for raw latent distance, while global effective rank does not explain planning. These results support a narrow conclusion: prediction can supply a useful state, but planning requires additional structure that orders action consequences.

## 1 INTRODUCTION

Language agents are commonly trained to predict which words come next, even when their task is to choose an action according to its consequences. We ask whether an agent can instead predict the latent consequence of each language action and plan with those predicted states. This question separates two capabilities that autoregressive generation usually entangles. A model must represent what has happened, and it must order the actions that remain.

Joint-embedding predictive architectures provide a natural substrate for the first capability (LeCun, 2022; Assran et al., 2023). An action-conditioned predictor maps the current representation and a language action to a predicted successor representation. Such a model can preserve consequence-relevant information without reconstructing every word. Action-conditioned JEPAs have already supported planning in physical environments (Assran et al., 2025). Language actions raise a focused question. If the next latent state is predicted accurately, does its geometry support choosing an action?

The answer is not automatic. Predictive learning can identify which histories have the same controlled future, but it need not identify a metric that ranks different futures by progress toward a goal. A bijective re-embedding preserves every latent transition and can still reverse the distances of two successor states to the goal. Prediction gives a state representation. It does not by itself give a planning geometry.

We study this gap in a controlled language-action testbed. Each stylized iGSM episode presents an arithmetic dependency problem. At each step, every learned method receives the same shuffled menu of executable intent phrases. The environment executes the selected phrase and returns its textual consequence. Supplying the current feasible menu is an intentional laboratory intervention. It holds

proposal and syntactic validity fixed so that the experiment compares how learning objectives value the same language actions.

We introduce Geometric Advantage Ranking (GAR). GAR uses factual and counterfactual action outcomes to impose local order constraints among candidate consequences. In the main JEPA, a scorer consumes the predicted successor. In a matched direct GAR control, the scorer consumes the history and action without using the predicted successor in its policy path. This terminology separates the supervision from the architectural factorization.

Our empirical goal is explanatory rather than competitive. A predictive GAR JEPA attains 43.3% strict success over five seeds. A token language model attains 42.2%, a sentence language model attains 47.1%, and a sentence model with an auxiliary latent objective attains 52.1%. The JEPA is therefore a viable planner under the shared interface, but it is not the best model. The more informative result is that latent prediction alone attains 18.3%. The full method attains 43.3%, and removing the GAR terms attains 7.6%. Counterfactual ordering is the main source of the gain, while counterfactual successor prediction provides an additional benefit.

The paper develops one connected argument.

1. We define the state needed for language-mediated control as a quotient of histories that have identical controlled futures, and show why this state is control sufficient.
2. We show that exact latent transition prediction does not identify a goal-directed metric without further assumptions. GAR supplies local counterfactual order constraints that are sufficient for strict planning when they recover the optimal action ordering.
3. We test the mechanism with five-seed objective ablations, gradient-routing controls, matched GAR scorers, and representation diagnostics. The evidence supports GAR as useful ordering supervision. It does not yet establish that successor factorization is uniformly better than direct scoring.

The resulting claim is deliberately narrow: action-conditioned latent prediction can support language-action planning, but predictive accuracy alone leaves the decision geometry underdetermined.

## 2 LANGUAGE-MEDIATED PREDICTIVE CONTROL

### 2.1 LANGUAGE AS A NON-CANONICAL INTERFACE

Consider a controlled process

$$\mathcal{M} = (\mathcal{S}, \mathcal{U}, P, \Omega, O, c, \mathcal{G}). \quad (1)$$

The semantic world state is  $s_t \in \mathcal{S}$  and the semantic intervention is  $u_t \in \mathcal{U}$ . The agent observes language  $x_t \in \Omega$  and chooses a language action  $a_t$ . A grounding map  $\alpha$  associates the phrase with an intervention,  $\alpha(a_t) = u_t$ . Language is non-canonical: different wording may express the same state or intervention, irrelevant facts may be present, and task-relevant facts may only become clear through history.

The agent’s information is therefore

$$h_t = (x_0, a_0, x_1, \dots, a_{t-1}, x_t). \quad (2)$$

Our experiments use a deterministic, finite-horizon specialization. The transition  $T(h, a)$  appends the environment’s deterministic consequence of  $a$  to  $h$ . This restriction matters because a point predictor in a general stochastic process can represent a conditional mean rather than a complete multimodal future.

### 2.2 PREDICTIVE STATE AS A QUOTIENT OF HISTORIES

Predictive state representations define state through action-conditioned predictions of future observations rather than recovery of a privileged hidden parameterization (Littman et al., 2001). We adapt that view to language-mediated control. Let  $Y$  contain future task-relevant observations and terminal outcomes,  $C$  future costs, and  $F$  future feasible-action sets. Two histories are *controlled-predictively*

*equivalent* when every admissible policy induces the same future law. The policies below range over grounded action continuations that are admissible from both histories:

$$h \sim_{\text{pred}} h' \iff \mathcal{L}_\pi(Y_{t+1:\tau}, C_{t:\tau}, F_{t:\tau} \mid h) = \mathcal{L}_\pi(Y_{t+1:\tau}, C_{t:\tau}, F_{t:\tau} \mid h') \quad \forall \pi. \quad (3)$$

In the deterministic testbed, this says that every semantically admissible action continuation has the same task-relevant consequences. The ideal predictive state is the equivalence class

$$q(h) = [h]_{\sim_{\text{pred}}} \in \mathcal{H}/\sim_{\text{pred}}. \quad (4)$$

This quotient formalizes the desired invariances. Paraphrased histories with identical controlled futures should meet. Negation and operator changes should remain separate when they change a consequence. An irrelevant entity rename may be removed. A representation that merges histories with different controlled futures is insufficient.

**Proposition 1** (Predictive equivalence is control sufficient). *If  $h \sim_{\text{pred}} h'$  includes equality of future costs and terminal events under every admissible policy, then  $V^\pi(h) = V^\pi(h')$  for every policy  $\pi$ . Hence  $V^*(h) = V^*(h')$ .*

*Proof.* The return is a measurable function of the future cost and terminal sequence. Equation 3 gives the same distribution of that sequence under each policy, and therefore the same expected return. Taking the optimum over policies preserves equality.  $\square$

The proposition identifies what a sufficient state must preserve. It is standard in spirit. Bisimulation gives a recursive local account through agreement in immediate reward and transitions between equivalence classes (Ferns et al., 2011). Our claim is weaker than learning a complete bisimulation metric. GAR constrains selected, goal-conditioned action orderings.

### 2.3 COMPRESSION IS A DESIDERATUM, NOT AN IMPLEMENTED OBJECTIVE

The quotient is the coarsest exact partition that preserves the controlled futures in Equation 3. This supplies a principled compression narrative without asserting that our neural model finds the minimal quotient. The information bottleneck seeks a compressed code of an input that retains information about a target (Tishby et al., 2000). A controlled predictive analogue would trade  $I(H; Z)$  against information about future outcomes conditional on future actions. Our method does not optimize such a rate term. Latent width and effective rank are not evidence of an explicit information bottleneck.

## 3 FROM PREDICTIVE STATE TO PLANNING GEOMETRY

### 3.1 PREDICTION LEAVES THE METRIC UNDERDETERMINED

Let an encoder  $\phi$  map a history to  $z_h$ , let  $e(a)$  encode a language action, and let  $F$  predict the deterministic successor:

$$F(\phi(h), e(a)) = \phi(T(h, a)). \quad (5)$$

**Proposition 2** (Exact prediction does not identify planning geometry). *Suppose  $(\phi, F)$  predicts every realized transition exactly. Let  $b$  be any bijection on the realized latent states. Define  $\tilde{\phi} = b \circ \phi$  and*

$$\tilde{F}(b(z), e(a)) = b(F(z, e(a))). \quad (6)$$

*Then  $(\tilde{\phi}, \tilde{F})$  is also prediction perfect. Unless  $b$  preserves the chosen metric, it need not preserve the order of successor distances to a goal. On a finite realized state set, a bijection can reverse the goal-distance ordering of two successors while retaining every transition.*

*Proof.* Substitution gives  $\tilde{F}(\tilde{\phi}(h), e(a)) = b(\phi(T(h, a))) = \tilde{\phi}(T(h, a))$ . For two distinct successor states and a goal, assign their images to any three distinct points. Choosing the first successor farther from the goal than the second reverses their metric order without changing the transition identity.  $\square$

Figure 1 shows the consequence. This proposition applies without extra identifiability assumptions or planning-aware constraints. Controlled world models can be identifiable up to more restricted maps, including orthogonal transformations, under strong structural and excitation assumptions (Zhang et al., 2026).

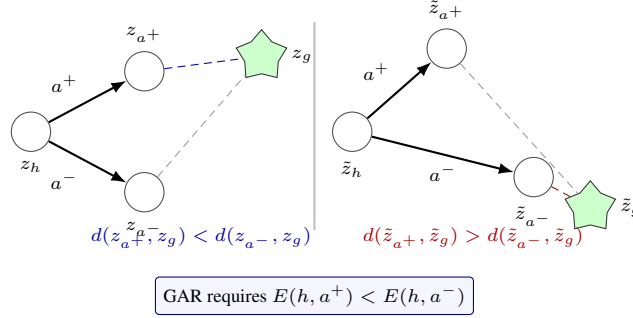


Figure 1: Transition prediction fixes which successor follows each action, but not how successors are arranged relative to the goal. Both embeddings have zero transition error. GAR adds an order constraint that excludes the right-hand arrangement for the illustrated preference.

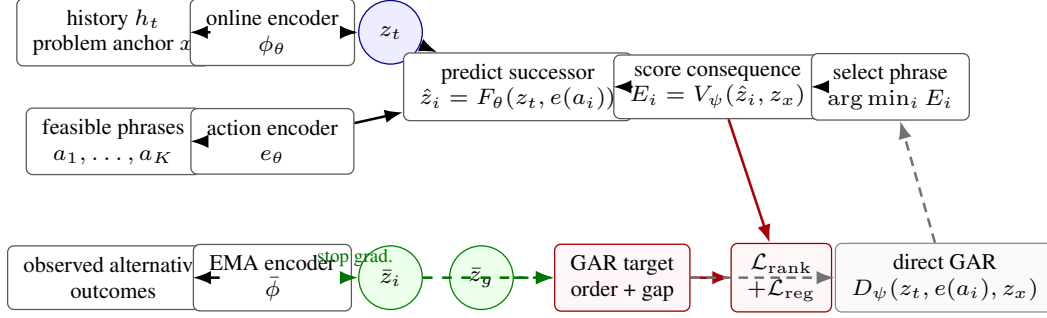


Figure 2: The predictive GAR factorization. The model encodes a language history, predicts one successor for each feasible phrase, and applies same-state GAR supervision. The transition-value realization scores predicted successors. A matched direct GAR realization bypasses the successor only in the policy path.

### 3.2 ACTION-CONDITIONED JOINT-EMBEDDING PREDICTION

The online encoder produces  $z_h = \phi_\theta(h)$  and the action encoder produces  $e_\theta(a)$ . The predictor gives

$$\hat{z}_{h,a} = F_\theta(z_h, e_\theta(a)). \quad (7)$$

An exponential-moving-average encoder produces a stopped-gradient target  $\bar{z}_{T(h,a)}$ . The factual and counterfactual prediction terms minimize normalized latent error for observed successor histories. The target encoder is always in evaluation mode and all dropout is disabled.

### 3.3 GEOMETRIC ADVANTAGE RANKING

Let  $z_g$  denote a target representation and define a latent potential  $\Phi_g(z) = -d(z, z_g)$ . The geometric progress of an action is

$$\Delta_\theta(h, a; g) = d(z_h, z_g) - d(\hat{z}_{h,a}, z_g). \quad (8)$$

For candidates  $a_i$  and  $a_j$ , GAR uses observed counterfactual outcomes to derive a stopped-gradient preference label. In the implemented geometric target, each true successor is encoded by the EMA network and compared with the episode’s terminal EMA state. The default horizon-two target extends this comparison with a bounded reference continuation. It is a geometric progress surrogate, not a Bellman-consistent advantage. No theorem guarantees that this surrogate orders the true control advantage. That alignment is an empirical hypothesis tested by the local action audits.

At deployment the terminal representation is unavailable. The prompt and query are encoded as a problem anchor  $z_x$ . Let  $E_\psi(h, a, x)$  be a lower-is-better action energy. The main transition-value realization is

$$E_\psi^{\text{TV}}(h, a, x) = V_\psi(\hat{z}_{h,a}, z_x). \quad (9)$$

The direct realization uses  $E_{\psi}^{\text{direct}}(h, a, x) = D_{\psi}(z_h, e(a), z_x)$  and does not consume  $\hat{z}_{h,a}$  in its policy path. Both are trained by GAR. Raw-distance GAR uses  $d(\hat{z}_{h,a}, z_g)$  and has no learned scoring head. This last diagnostic receives the encoded terminal state and is not a deployable policy.

For each pair whose target distances satisfy  $\bar{d}_i + \epsilon < \bar{d}_j$ , the ranking loss is

$$\mathcal{L}_{\text{rank}} = \sum_{i>j} [m + E_{\psi}(h, a_i, x) - E_{\psi}(h, a_j, x)]_+. \quad (10)$$

A pairwise regression term calibrates energy differences to target-distance differences:

$$\mathcal{L}_{\text{reg}} = \sum_{i<j} [(E_j - E_i) - \beta(\bar{d}_j - \bar{d}_i)]^2. \quad (11)$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{factual}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \quad (12)$$

The default transition-value model allows the GAR gradients to reach the scorer, predictor, action encoder, and online state encoder. The target geometry changes only through slow EMA tracking. GAR is related to potential-based reward shaping (Ng et al., 1999), but it does not add a fixed shaping reward to a known MDP. It jointly learns the representation, transition model, and action energy.

### 3.4 WHY LOCAL ORDERING IS ENOUGH

Let  $D^*(h)$  be the minimum remaining action count in a deterministic unit-cost puzzle. Define

$$A^*(h, a) = D^*(h) - 1 - D^*(T(h, a)). \quad (13)$$

An action on a shortest solution has value zero. An action that consumes a step without reducing the remaining distance has value  $-1$ .

**Proposition 3** (Local ordering suffices for strict planning). *If a learned score ranks every optimal action above every suboptimal action at every reachable history, greedy selection reaches the goal in exactly  $D^*(h_0)$  actions.*

*Proof.* At a history with distance  $d > 0$ , the selected action is optimal and reaches a history with distance  $d - 1$ . Repeating this argument reaches distance zero after exactly  $d$  actions.  $\square$

**Corollary 1** (Robust ordering). *If the smallest true action gap is  $\gamma > 0$  and  $\sup_{h,a} |\hat{A}(h, a) - A^*(h, a)| < \gamma/2$ , greedy selection preserves the optimal ordering.*

GAR does not receive  $A^*$  in the main experiment. The proposition states the control condition that its geometric surrogate should approximate. It predicts that same-state ranking accuracy and margin should track strict success more closely than global effective rank or one-step prediction error.

## 4 CONTROLLED EXPERIMENTS

### 4.1 A LABORATORY FOR CONDITIONAL PLANNING

We use a stylized iGSM generator with named quantities, arithmetic dependencies, distractors, and a target query (Ye et al., 2025). Training instances are generated online. The paper models see 300,000 examples with solution lengths from three to nine actions. Each completed headline result averages five seeds, and each seed is evaluated on 200 fresh episodes.

At a decision point, the environment supplies a shuffled set of executable intent phrases. It does not mark necessary actions, reveal their outcomes, or expose future menus. Every learned policy receives the same current set. This interface is candidate privileged relative to unrestricted language generation. It is appropriate for the present question because it removes proposal recall and malformed actions from the comparison. The claim is conditional planning among executable language actions.

Strict success requires solving a puzzle in the minimum number of actions. Success with  $k$  excess actions permits at most  $k$  additional decisions. We report strict and  $+2$  results for the completed suite. The final evaluation will add the full budget curve at  $k \in \{0, 1, 2, 4\}$ , invalid-action rate for unrestricted controls, and paired confidence intervals.

## 4.2 MODELS

The main JEPA uses 256-dimensional history and action representations with an MLP successor predictor. The causal JEPA replaces the successor predictor with a causal Transformer over encoded history states. The token LM scores each candidate by its token likelihood. The sentence LM predicts the next intent phrase as a sentence-level target. The sentence-plus-latent model adds next-sentence latent prediction to the sentence objective. All policies select from the same shuffled current menu.

The main GAR realization is the transition-value factorization in Section 3. We compare it with factual and counterfactual latent prediction without GAR, ranking without regression, GAR without counterfactual latent prediction, detached GAR gradients, raw successor distance, and direct GAR. A shuffled-label control preserves the candidate sets and counterfactual transitions while destroying only the within-state quality correspondence.

## 4.3 TUNING AND UNCERTAINTY

Completed learned headline and primary ablation rows use five training seeds. Hyperparameters are chosen on validation data. The present tables report mean and sample standard deviation across seed-level success rates. The random policy is not yet reported because it has not been rerun on the paired headline episodes. Before submission, headline claims will use identical puzzle indices across methods, more validation episodes per seed, paired bootstrap intervals, and paired intervals for model differences.

Each model-dataset pair receives its own learning-rate search budget. Ablations reuse the selected rate unless they change loss scale or gradient routing. Such changes receive a local cross-check. No test result selects a hyperparameter.

## 4.4 FOUR EMPIRICAL QUESTIONS

**RQ1: Can latent prediction support planning?** We compare predictive and generative candidate scorers under the common menu.

**RQ2: Is prediction sufficient?** We remove GAR, remove successor losses, vary the ranking and regression terms, vary the number of counterfactuals, and vary the target horizon.

**RQ3: Does GAR create decision-relevant geometry?** At aligned histories, we measure pairwise feasible-action ranking, top-1 optimal recall, mean regret, rank correlation with exact continuation cost, and the optimal-versus-best-suboptimal margin. Exact continuation cost is used only for evaluation. We compare transition-value GAR, raw distance, detached gradients, and direct GAR.

**RQ4: What is preserved and discarded?** Frozen-feature probes measure operation identity, remaining actions, resolved quantities, and numerical outcomes. Effective rank checks collapse. Controlled paraphrase, negation, operator-swap, and entity-renaming interventions are required to test the quotient account directly.

## 4.5 EVIDENCE BOUNDARY

Only stylized iGSM has a complete five-seed headline comparison. Faithful iGSM and non-arithmetic domains remain outside the current main claims. Test-time scaling is also outside the evidence table. The existing deep-rollout evaluator and FLOP accounting do not yet support an interpretable comparison. We will include scaling only after fixed-work evaluation, stable recursive prediction, and matched recurrent baselines pass their validity checks.

# 5 CAN LATENT PREDICTION PLAN?

## 5.1 VIABILITY UNDER A COMMON ACTION INTERFACE

Table 1 answers RQ1. The MLP JEPA with GAR attains .433 strict success. Its mean is close to the token LM at .422, below the sentence LM at .471, and below the sentence-plus-latent model at .521.

Table 1: Closed-loop planning on stylized iGSM. Values are mean  $\pm$  sample standard deviation over five seeds. Every method scores the same shuffled current feasible-action menu.

| Model                | Strict success                    | Success with +2 actions           |
|----------------------|-----------------------------------|-----------------------------------|
| Sentence LM + latent | <b>.521 <math>\pm</math> .047</b> | .809 $\pm$ .020                   |
| Sentence LM          | .471 $\pm$ .046                   | .813 $\pm$ .023                   |
| MLP JEPA + GAR       | .433 $\pm$ .019                   | .796 $\pm$ .033                   |
| Token LM             | .422 $\pm$ .124                   | <b>.836 <math>\pm</math> .073</b> |
| Causal JEPA + GAR    | .366 $\pm$ .047                   | .778 $\pm$ .043                   |
| Random policy        | <i>pending</i>                    | <i>pending</i>                    |

Table 2: Five-seed mechanism ablations. Rank is the GAR pairwise hinge. Reg. calibrates pairwise energy differences to geometric target differences. CF pred. predicts counterfactual successor states.

| Configuration             | Rank | Reg. | CF pred. | Strict          | +2              |
|---------------------------|------|------|----------|-----------------|-----------------|
| Full predictive GAR       | yes  | yes  | yes      | .433 $\pm$ .019 | .796 $\pm$ .033 |
| GAR ranking only          | yes  | no   | yes      | .358 $\pm$ .045 | .750 $\pm$ .018 |
| GAR without CF prediction | yes  | yes  | no       | .345 $\pm$ .049 | .734 $\pm$ .017 |
| Latent prediction only    | no   | no   | yes      | .183 $\pm$ .016 | .628 $\pm$ .031 |
| Factual prediction only   | no   | no   | no       | .076 $\pm$ .013 | .422 $\pm$ .022 |

The causal JEPA attains .366. The MLP JEPA has the smallest seed standard deviation among the learned methods, although five seeds do not establish a general stability advantage.

The result establishes that a non-generative latent consequence model can support nontrivial closed-loop planning. It does not establish that JEPA is better than language modeling. The leading sentence-plus-latent row instead suggests that generative and predictive objectives can provide complementary biases.

## 5.2 PREDICTION ALONE IS NOT SUFFICIENT

Table 2 gives the central empirical result. Factual and counterfactual latent MSE without GAR attains .183 strict success. The full predictive GAR model attains .433. A factual-prediction-only configuration, labeled no GAR in the run ledger, attains .076. Ranking without pairwise regression attains .358. Removing the counterfactual latent loss while retaining GAR attains .345.

These rows support two conclusions. Same-state action ordering accounts for the larger part of the planning gain. Counterfactual successor prediction adds information beyond the ordering labels. The full gain is not explained by additional alternative transitions alone.

The target horizon is not monotonic. Horizons one, two, and four attain .161, .433, and .181 strict success. This is a property of the present geometric reference-continuation target, not a general claim that two-step supervision is optimal. Increasing counterfactual breadth from one to four changes strict success from .403 to .439. Pairwise-regression weights .10 and .50 attain .443 and .402. Nearby coefficients therefore preserve much of the effect.

## 5.3 INFORMATION-MATCHED CONTROLS

Shuffling the geometric quality labels while retaining counterfactual transitions gives .015 strict success at each of three screened learning rates. A direct GAR scorer without successor auxiliaries reaches .150 to .185. The same direct GAR policy path with JEPA successor auxiliaries reaches .440 in one seed, compared with .435 for transition-value GAR in that seed. These controls show that the useful signal is not simply the number of counterfactual examples. They also prevent a stronger conclusion. In-distribution performance does not yet show that routing the policy through a predicted successor is better than direct GAR scoring.

The appropriate next test is not another leaderboard comparison. It is a matched comparison under data scarcity, longer compositions, action paraphrases, altered goals, and stable multi-step simulation.

Table 3: Five-seed frozen-feature diagnostics. Probe scores are means over seeds. Effective rank measures non-collapse. It does not track strict planning performance.

| Model                | Eff. rank | Operation bal. acc. | Remaining $R^2$ | Resolved $R^2$ | Outcome $R^2$ |
|----------------------|-----------|---------------------|-----------------|----------------|---------------|
| MLP JEPA             | 198.70    | .406                | .346            | .895           | .088          |
| Causal JEPA          | 215.34    | .403                | .332            | .846           | .053          |
| Token LM             | 3.55      | .532                | .439            | .986           | -.020         |
| Sentence LM          | 79.64     | .777                | .410            | .879           | -.027         |
| Sentence LM + latent | 71.44     | .807                | .429            | .886           | -.029         |

A benefit on one of these axes would support the predictive factorization rather than GAR supervision alone.

## 6 WHAT GEOMETRY IS LEARNED?

### 6.1 THE CURRENT EVIDENCE IS LOCAL AND INCOMPLETE

GAR makes a local action-ordering claim. In a one-seed aligned-state audit, the deployed transition-value energy attains .826 pairwise ranking accuracy and .786 top-1 optimal-action recall. Raw predicted successor distance attains .603 and .549. The learned energy is therefore more decision useful than uncalibrated latent distance in this checkpoint.

This result does not yet prove that GAR reshapes the representation. The transition-value head may learn a supervised readout while the underlying geometry changes little. The one-seed gradient-routing controls are consistent with both effects. Full predictive GAR reaches .435 strict success, while detaching GAR gradients from the predictive body reaches .375. Raw-distance selection reaches .190. Direct GAR with successor auxiliaries reaches .440. Five-seed completion and checkpoint-level local metrics are needed before attributing the gap to geometry shaping.

The decisive analysis will measure pairwise ranking accuracy, top-1 recall, mean regret, Spearman correlation with exact continuation cost, and the optimal-versus-best-suboptimal margin for every retained checkpoint and seed. It will then correlate each quantity with strict success. The theoretical account predicts that local ranking and margin will track planning more closely than one-step latent error, successor retrieval, effective rank, or global probe scores.

### 6.2 GLOBAL RANK IS NOT A PLANNING EXPLANATION

Table 3 reports the completed frozen-feature diagnostics. Both JEPA spaces are non-collapsed, with effective ranks near 200 in 256 dimensions. The token LM has effective rank 3.55 and remains competitive in planning. Effective rank is therefore a health check, not a control metric.

Sentence representations make operation identity much more linearly accessible than the JEPA states. MLP JEPA strongly exposes resolved-variable count. That variable is confounded with elapsed step index, so it cannot yet be interpreted as semantic progress. Remaining distance is moderately decodable across all systems. Numerical outcomes are weakly decoded. The existing necessary-action probe is at chance and is not used as positive evidence.

### 6.3 A PROGRESS-PLUS-RESIDUAL HYPOTHESIS

A representation may decompose as

$$z = z_{\text{progress}} + z_{\text{residual}}. \quad (14)$$

A small goal-relative component can govern local action ordering while many tangential directions retain predictive detail. This would reconcile high JEPA rank with a simple planning readout. We will test the hypothesis by predicting remaining distance from the scalar goal distance, the GAR energy, the full state, and the residual after removing the estimated goal-progress direction. The hypothesis is supported only if removing that direction selectively damages action ordering while leaving other predictive probes intact.



## 6.4 CONTROLLED SEMANTIC INTERVENTIONS

An ordinary t-SNE or UMAP cloud cannot establish the quotient account. The required plot must use controlled pairs. We will encode identical semantic states under paraphrase, identical actions under paraphrase, high-overlap negations with changed consequences, operator swaps, irrelevant entity renaming, and histories matched on step index but not true progress. Full-space retrieval, pairwise distance, and action-order preservation are primary. A shared UMAP and t-SNE projection will illustrate the same frozen points and use identical hyperparameters across JEPA and LM representations.

These paired features have not yet been collected. We therefore omit a manifold plot rather than presenting an uncontrolled projection as evidence. The required outcome is clear: semantic paraphrases should retrieve one another, while a negation should separate only when it changes the controlled future. Failure on either side would contradict the quotient interpretation.

## 7 RELATED WORK

**Predictive state and control equivalence.** Predictive state representations describe state through multi-step action-conditioned predictions of future observations (Littman et al., 2001). Bisimulation identifies states that agree in reward and transition behavior, while approximate bisimulation metrics relate representation distance to value continuity (Ferns et al., 2011). We use these ideas to define the control-sufficient quotient over language histories. We do not claim to learn the complete quotient or a full bisimulation metric.

**Joint-embedding prediction.** JEPA proposes learning predictive representations without reconstructing all observation detail (LeCun, 2022). I-JEPA and V-JEPA instantiate feature prediction for images and video (Assran et al., 2023; Bardes et al., 2024). V-JEPA 2 adds an action-conditioned model for robot planning toward image goals (Assran et al., 2025). Language-focused variants include LLM-JEPA and JEPA-Reasoner (Huang et al., 2026; Liu et al., 2026). Our subject is controlled language-action consequence prediction and same-state counterfactual action ordering.

**Planning-aware predictive geometry.** Recent methods also address the gap between predictive representations and planning. Value-guided JEPA trains goal-conditioned distance with a value-learning objective (Destrade et al., 2026). Temporal straightening regularizes latent trajectory curvature (Wang et al., 2026). Temporal-Distance JEPA learns a directed temporal cost from trajectory order, cross-trajectory negatives, and rollout consistency (Bai & Xiong, 2026). GAR instead imposes within-state order constraints over alternative discrete language actions. Its transition-value realization distills that ordering into a scorer over predicted successors. Direct GAR tests whether the predictive factorization adds value beyond the same supervision.

The overlap changes the novelty claim. This is not the first JEPA planner and not the first planning-aware latent geometry. The contribution is the action-level counterfactual formulation, its control-sufficiency account, and a controlled comparison of predictive and direct candidate scoring in language.

**Latent world models and shaping.** Dreamer and MuZero show that latent dynamics become decision useful when policy, reward, or value objectives shape their representations (Hafner et al., 2020; Schrittwieser et al., 2020). Potential-based shaping adds a fixed potential difference while preserving optimal policies under stated conditions (Ng et al., 1999). GAR learns its representation, geometric targets, and energy jointly, so it is related to shaping but not an instance of the classical fixed-potential result.

**Information bottleneck and test-time computation.** The information bottleneck formalizes compression that preserves target-relevant information (Tishby et al., 2000). We use it only as a related desideratum because the present model has no explicit rate penalty. More inference computation can improve reasoning through search, sampling, or recurrent depth (Snell et al., 2024; Geiping et al., 2025). Latent rollout offers a promising route, but the current project lacks a valid FLOP-matched scaling result. We make no empirical scaling claim in this draft.

## 8 CONCLUSION

Predictive learning and planning impose different constraints on a representation. Controlled-future equivalence identifies which language histories may share a state. Transition prediction preserves how actions move between those states. Neither condition fixes a goal-directed metric. GAR adds the missing local order supervision by comparing counterfactual action consequences.

The experiments support this mechanism-level account. A non-generative JEPA is a viable conditional planner, although the strongest sentence model performs better. Latent prediction alone is substantially weaker than predictive GAR. Counterfactual successor prediction and counterfactual ordering are complementary. Raw distance is weaker than the learned GAR energy, while a matched direct GAR scorer remains competitive in the current in-distribution test.

The final point is as important as the positive result. The evidence supports GAR supervision, but does not yet show that successor factorization is uniquely valuable. Controlled semantic interventions, local geometry correlations, matched generalization tests, a non-arithmetic domain, and a valid test-time scaling protocol determine whether the stronger representation claim survives.

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## A LIMITATIONS AND SCOPE

The main interface supplies the current symbolic feasible-action menu. This isolates action selection but does not test unrestricted proposal or language generation. The results establish conditional planning among executable phrases. They do not establish an autonomous language agent.

The completed evidence comes from a stylized arithmetic environment. Templated language and exact counterfactual transitions are valuable for mechanism analysis, but they limit external validity. Faithful iGSM and at least one non-arithmetic domain are required before submission.

Several mechanism controls use one seed. Direct GAR currently matches transition-value GAR in that setting. Detached gradients, raw-distance planning, local metric correlations, and controlled semantic interventions need five-seed completion before they support a central claim.

The current deeper-planning protocol does not support a test-time-scaling claim. A valid result requires a shared proposal interface or exhaustive catalogue search, paired fixed-work evaluation, stable multi-step prediction, and calibrated FLOPs for recurrent baselines. We omit the current curves because they do not isolate the intended effect of additional inference computation.

The present model is trained from scratch on a controlled language. Its findings do not directly establish how pretrained large language models would organize open-domain planning states. Conversely, the controlled setting enables exact causal comparisons that would be difficult to interpret in an open-ended agent benchmark.

## B DETAILED PROTOCOL

**Information boundary.** Headline policies observe the problem statement, prior selected intent phrases and outcomes, and a shuffled menu of actions executable at the current state. They do not observe the expert action, exact remaining distance, future action availability, terminal state, or counterfactual outcomes at evaluation. Training uses exact alternative outcomes only for declared counterfactual prediction and GAR targets.

**Target encoder.** The EMA encoder is updated from the online encoder and kept in evaluation mode at all times. Gradients do not enter EMA targets. Dropout is disabled in the encoder, predictor, action encoder, scorer, and target encoder. These conditions follow earlier validity tests that found stochastic or train-mode targets destabilize dense rollout supervision.

**Paired evaluation.** Final comparisons will use identical generated puzzle indices, action-order seeds, and budgets across systems. We will report per-seed means, sample standard deviations, paired bootstrap intervals, and paired differences for headline claims. The currently completed five-seed results use 200 fresh validation episodes per seed. The fixed-checkpoint scaling gate uses 1,000 paired episodes.

**Test-time scaling admission.** A scaling figure will enter the paper only when every point uses the same paired puzzles and a fixed, measured work definition. Candidate proposal, branching, and terminal handling must not expose solution status. Recursive transition error and action ranking must be audited at each depth. Recurrent LM baselines must use the same candidate work and independently measured FLOPs. The existing pilot does not meet the complete admission rule and is excluded.

**Tuning contract.** For the final suite, width-256 learning rates are screened on validation and reused within a model-dataset pair. The planned grid is  $\{5e-5, 7e-5, 10^{-4}, 3e-4, 5e-4, 7e-4, 10^{-3}, 3e-3, 5e-3\}$ . The initial stage uses a compact subset. Five seeds are trained only after selecting the validation learning rate. Ablations reuse that rate unless they change gradient routing or loss scale.

**Why current and historical numbers differ.** Historical reports used different generators, predictor variants, batch sizes, candidate interfaces, and training budgets. This draft includes only the 300,000-example paper family and matched controls retrieved from its run lineage. Older values are retained in the repository but are not combined with the current table.

## C COMPLETE CURRENT RESULT TABLES

Table 4: Five-seed GAR ablations on stylized iGSM.

| Variant                       | Strict          | +2 actions      |
|-------------------------------|-----------------|-----------------|
| Full GAR                      | .433 $\pm$ .019 | .796 $\pm$ .033 |
| No GAR                        | .076 $\pm$ .013 | .422 $\pm$ .022 |
| Latent MSE only               | .183 $\pm$ .016 | .628 $\pm$ .031 |
| Ranking only                  | .358 $\pm$ .045 | .750 $\pm$ .018 |
| No counterfactual latent loss | .345 $\pm$ .049 | .734 $\pm$ .017 |
| GAR horizon 1                 | .161 $\pm$ .021 | .597 $\pm$ .037 |
| GAR horizon 2                 | .433 $\pm$ .019 | .796 $\pm$ .033 |
| GAR horizon 4                 | .181 $\pm$ .008 | .651 $\pm$ .014 |
| One counterfactual            | .403 $\pm$ .041 | .772 $\pm$ .020 |
| Four counterfactuals          | .439 $\pm$ .049 | .821 $\pm$ .025 |
| Advantage MSE weight .10      | .443 $\pm$ .037 | .802 $\pm$ .034 |
| Advantage MSE weight .50      | .402 $\pm$ .039 | .783 $\pm$ .026 |

## D PREDECLARED ANALYSES AND DECISION RULES

**Local planning geometry.** For every feasible action pair at aligned histories, report pairwise ranking accuracy, top-1 optimal recall, mean action regret, Spearman correlation with exact continuation cost, and the optimal-versus-best-suboptimal margin. Across seeds and retained checkpoints, correlate each quantity with strict planning success. Compare those correlations with one-step latent error, successor retrieval, effective rank, and global probes. The geometry account is supported only if local ordering and margin track success materially better.

Table 5: One-seed mechanism controls. These rows are exploratory. Raw distance uses an oracle-terminal embedding and is candidate privileged. GAR names the supervision. TV and direct name the scoring factorization.

| Variant                                | Strict | +2 actions |
|----------------------------------------|--------|------------|
| Transition-value GAR                   | .435   | .820       |
| Detached GAR body                      | .375   | .775       |
| GAR without counterfactual latent loss | .255   | .695       |
| Counterfactual latent only             | .080   | .410       |
| Direct GAR with successor auxiliaries  | .440   | .820       |
| Raw successor distance <sup>†</sup>    | .190   | .625       |

Table 6: Five-seed representation diagnostics. Probe values are means across seeds.

| Model                | Effective rank | Operation bal. acc. | Remaining $R^2$ | Resolved $R^2$ | Outcome $R^2$ |
|----------------------|----------------|---------------------|-----------------|----------------|---------------|
| MLP JEPA             | 198.70         | .406                | .346            | .895           | .088          |
| Causal JEPA          | 215.34         | .403                | .332            | .846           | .053          |
| Token LM             | 3.55           | .532                | .439            | .986           | -.020         |
| Sentence LM          | 79.64          | .777                | .410            | .879           | -.027         |
| Sentence LM + latent | 71.44          | .807                | .429            | .886           | -.029         |

**Gradient routing and factorization.** Compare transition-value GAR, detached-body GAR, raw-distance GAR, and direct GAR. All learned scorers receive the same histories, candidate phrases, counterfactual outcomes, ordering labels, and tuning budget. Full geometry shaping requires a gain over detached GAR in head-free local geometry. Predictive factorization requires a gain over direct GAR on at least one preregistered axis: data efficiency, compositional length, action paraphrase, altered goals, forced-error recovery, or valid multi-step simulation.

**Counterfactual information.** Compare full predictive GAR, counterfactual successor prediction without GAR, shuffled GAR labels, direct GAR without successor auxiliaries, and direct GAR with the same auxiliaries. This factorial separates alternative-transition coverage, meaningful quality labels, auxiliary representation learning, and successor-mediated scoring.

**Controlled semantic geometry.** Construct paired histories for state paraphrase, action paraphrase, negation, operator swap, irrelevant entity renaming, and matched elapsed step with different true progress. Report full-space retrieval, normalized pairwise distances, and action-order preservation before any projection. Produce t-SNE and UMAP from the same frozen point set with shared hyperparameters across models. The projections are illustrations, not primary statistics.

**Progress and residual structure.** Predict remaining distance from raw goal distance, GAR energy, the full latent state, and the residual after removing the estimated progress direction. A radial-progress account requires local action ordering to degrade after removal while successor retrieval and non-progress probes remain comparatively intact.

**Why sentence models lead.** At aligned histories, decompose errors into candidate ranking, recovery after a forced distractor, action-type calibration, and lexical-template sensitivity. Score the sentence-plus-latent model through both its generative readout and its latent readout. This localizes whether its advantage lies in state encoding, next-phrase generation, or their combination.

**Test-time computation.** The scaling experiment is admitted only after a fixed-work evaluator, stable recursive transition audit, and calibrated recurrent baselines pass. Report every tested compute point on paired puzzles. A decrease with depth must first be localized to transition drift, score calibration, proposal breadth, or terminal handling. It is not presented as a test-time-scaling result by itself.

Table 7: One-seed information controls across a local learning-rate screen. These controls separate meaningful counterfactual ordering from counterfactual exposure alone.

| Control                                  | Learning rate      | Strict | +2 actions |
|------------------------------------------|--------------------|--------|------------|
| Direct GAR without successor auxiliaries | $10^{-4}$          | .180   | .590       |
| Direct GAR without successor auxiliaries | $3 \times 10^{-4}$ | .185   | .635       |
| Direct GAR without successor auxiliaries | $7 \times 10^{-4}$ | .150   | .560       |
| GAR with shuffled quality labels         | $10^{-4}$          | .015   | .170       |
| GAR with shuffled quality labels         | $3 \times 10^{-4}$ | .015   | .175       |
| GAR with shuffled quality labels         | $7 \times 10^{-4}$ | .015   | .175       |

Table 8: Paper-suite completion matrix. Empty entries are intentional.

| Experiment family        | Stylized iGSM  | Faithful iGSM  | ProofWriter    | Blocksworld / ALFWorld |
|--------------------------|----------------|----------------|----------------|------------------------|
| Five-seed headline       | complete       | <i>pending</i> | <i>pending</i> | <i>pending</i>         |
| Five-seed GAR ablation   | complete       | <i>pending</i> | <i>pending</i> | <i>pending</i>         |
| Local ordering audit     | one seed       | <i>pending</i> | <i>pending</i> | <i>pending</i>         |
| Length OOD               | partial        | <i>pending</i> | <i>pending</i> | <i>pending</i>         |
| Controlled minimal pairs | <i>pending</i> | <i>pending</i> | <i>pending</i> | <i>pending</i>         |
| Test-time FLOP scaling   | invalid pilot  | <i>pending</i> | <i>pending</i> | <i>pending</i>         |

## E REPRODUCIBILITY AND ARTIFACT STATUS

The conceptual figures are repository-native TikZ and compile directly with the manuscript. Empirical results are transcribed from the retrieved current-findings ledger and linked to run identifiers in the paper README. No t-SNE, UMAP, or test-time scaling figure appears before its analysis protocol is valid and its source features are retained.

The completed runs use immutable code snapshots and checkpoint hashes. Failed and superseded jobs remain visible in the controller ledger. The manuscript excludes the current scaling pilots from evidence. The final artifact will provide a machine-readable manifest with checkpoint path, hash, model configuration, seed, validation split, evaluation indices, parameter count, optimizer updates, wall time, and measured FLOPs for every included row.

The official ICLR 2027 style files are vendored unchanged in the paper directory. Run `make figures` through `make_figures.sh`, then compile `main.tex` with `latexmk -pdf`. The draft remains anonymous because `\iclrfinalcopy` is disabled.